# PROJECT REPORT

## TITLE PAGE

**PROJECT TITLE : AI SECURITY LOG INTELLIGENCE PLATFORM**

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# ABSTRACT

**The AI Security Log Intelligence Platform is a web-based application designed to analyze and interpret system security logs using artificial intelligence. The system supports multiple types of logs including Windows, Linux, Application, and Web Server logs.**

**The platform detects potential security threats such as brute force attacks, suspicious login attempts, and malicious activities by analyzing log data. It provides detailed results including attack type, severity level, and supporting evidence.**

**An integrated AI chatbot enhances the system by automatically explaining detected threats and answering user queries related to cybersecurity. The chatbot uses a large language model to provide real-time insights and prevention strategies.**

**This project helps users understand security incidents more effectively and improves decision-making in cybersecurity environments.**

# INTRODUCTION

In today's digital world, cybersecurity plays a critical role in protecting systems and data. Organizations generate large amounts of log data, which contain valuable information about system activities and potential threats.

However, manually analyzing these logs is time-consuming and requires expertise. This project aims to automate the process of log analysis and make it more intelligent using artificial intelligence.

The AI Security Log Intelligence Platform allows users to upload different types of logs and receive automated analysis results. It also includes an AI assistant that explains threats and provides recommendations in a simple and understandable way.

# PROBLEM STATEMENT

Analyzing security logs manually is complex, time-consuming, and requires technical expertise. Many users find it difficult to understand the meaning of log entries and identify potential threats.

There is a need for a system that can automatically analyze logs, detect threats, and provide clear explanations in a user-friendly manner.

# OBJECTIVES

• To develop a system for analyzing different types of security logs

• To detect potential cyber attacks from log data

• To classify threats based on severity

• To provide evidence for detected attacks

• To integrate AI for automatic explanation and user interaction

• To create a user-friendly interface for easy access

# SYSTEM ARCHITECTURE

The system follows a client-server architecture.

The frontend is developed using HTML, CSS, and JavaScript, which provides an interactive user interface. The backend is built using Flask (Python), which handles user requests, processes data, and communicates with the database and AI services.

The system workflow is as follows:

1. The user logs into the system.

2. The user uploads a log file (Windows, Linux, Application, or Web logs)

**.**

**3. The backend processes the log using custom analyzers.**

**4. The system identifies potential threats and generates results including attack type, severity, and evidence.**

**5. The results are displayed to the user.**

**6. The user can click "Explain with AI" to open the chatbot.**

**7. The chatbot uses an AI model (via OpenRouter API) to explain the detected attack and answer user queries.**

**The system ensures efficient processing and provides real-time insights into security threats.**

# TECHNOLOGIES USED

**The following technologies were used in the development of this project:**

**• Python – Backend logic and log analysis**

**• Flask – Web framework for handling requests and routing**

**• HTML – Structure of web pages**

**• CSS – Styling and UI design**

**• JavaScript – Client-side interactivity and chatbot communication**

**• MySQL – Database for storing user credentials**

**• OpenRouter API – Integration of AI model for chatbot functionality**

**• JSON – Data exchange format between frontend and backend**

# METHODOLOGY

**The system follows a step-by-step methodology for analyzing logs and generating insights.**

**Step 1: User Authentication**
**Users register and log in securely using session-based authentication.**

**Step 2: Log Upload**
**Users upload log files of different types such as Windows, Linux, Application, or Web server logs.**

**Step 3: Log Analysis**
**The uploaded logs are processed using custom Python analyzers to detect suspicious patterns and activities.**

**Step 4: Threat Detection**
**The system identifies potential attacks and assigns severity levels based on the analysis.**

**Step 5: Result Generation**
**The results are displayed in a structured format including attack type, severity, and evidence.**

**Step 6: AI Integration**
**The chatbot automatically explains detected threats and allows users to ask additional questions.**

**Step 7: User Interaction**
**Users interact with the chatbot to gain deeper understanding and prevention strategies.**

# MODULES DESCRIPTION

**The system is divided into multiple modules:**

**1. User Authentication Module**
**Handles user registration and login using secure password hashing.**

**2. Dashboard Module**
**Provides access to different log analysis features.**

**3. Log Upload Module**
**Allows users to upload log files for analysis.**

**4. Log Analyzer Module**
**Processes logs and detects potential threats using predefined logic.**

**5. Result Module**
**Displays detected attacks, severity, and evidence in a structured format.**

**6. AI Chatbot Module**

**Provides intelligent explanations of detected threats and answers user queries using AI.**

**7. Database Module**
**Stores user data securely using MySQL.**

# IMPLEMENTATION DETAILS

**The system is implemented using Python with the Flask framework for backend processing. The frontend is developed using HTML, CSS, and JavaScript to provide an interactive user interface.**

**The application includes session-based authentication where user credentials are securely stored in a MySQL database using password hashing.**

**Log files are uploaded through the web interface and processed using custom-built analyzers written in Python. These analyzers identify patterns such as failed login attempts, suspicious requests, and abnormal activities.**

**The results are displayed in a structured format showing attack type, severity, and evidence.**

**The AI chatbot is integrated using the OpenRouter API, which connects to a large language model. The chatbot receives user queries and system-generated attack data, and provides real-time explanations and security recommendations.**

**The system ensures smooth communication between frontend and backend using JSON-based API requests.**

# RESULTS & OUTPUT

**The system successfully analyzes different types of security logs and detects potential threats.**

**The output includes:**
- **Attack Type (e.g., SQL Injection, Brute Force)**
- **Severity Level (Low, Medium, High)**
- **Evidence supporting the detection**

**The chatbot automatically explains detected threats and provides additional insights when users ask questions.**

**The system provides accurate and fast analysis, making it easier for users to understand security incidents without deep technical knowledge.**

# ADVANTAGES

- **Automated log analysis reduces manual effort**

- **Supports multiple types of logs (Windows, Linux, Application, Web)**

- **Provides clear and structured results**

- **AI-powered chatbot enhances understanding of threats**

**• User-friendly interface**
**• Fast and efficient processing**

**• Helps in quick decision-making for security issues**

# FUTURE SCOPE

**The system can be further enhanced with additional features such as:**

**• Real-time log monitoring instead of file upload**

**• Integration with SIEM tools**

**• Advanced machine learning models for better threat detection**

**• User-specific dashboards with analytics**

**• Alert system for critical threats**

**• Deployment on cloud platforms for scalability**

**• Voice-enabled AI assistant**

# CONCLUSION

**The AI Security Log Intelligence Platform provides an efficient solution for analyzing and understanding security logs. By combining log analysis with artificial intelligence, the system simplifies the process of detecting and explaining cyber threats.**

**The project demonstrates how AI can be used to enhance cybersecurity systems and improve user understanding. It serves as a valuable tool for both beginners and professionals in the field of information security.**